

MSPM0_K210 Barcode Recognition

1. K210 and MSPM0 Communication

1.1 Experiment Prerequisites

This tutorial uses the MSPM0G3519 development board. K210 needs to run the program in **K210-AI(MSPM0)** to start the experiment.

MSPM0G3519 *1

K210 vision module *1 (with SD card (with AI model), camera)

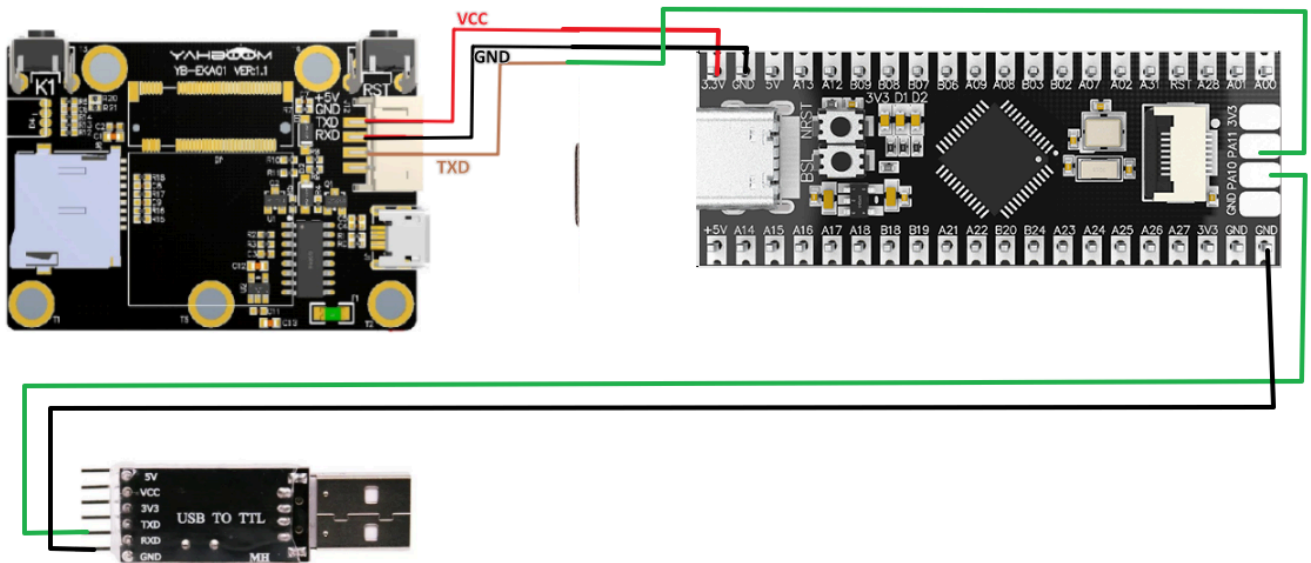
USB to TTL module *1

1.2 Experiment Wiring

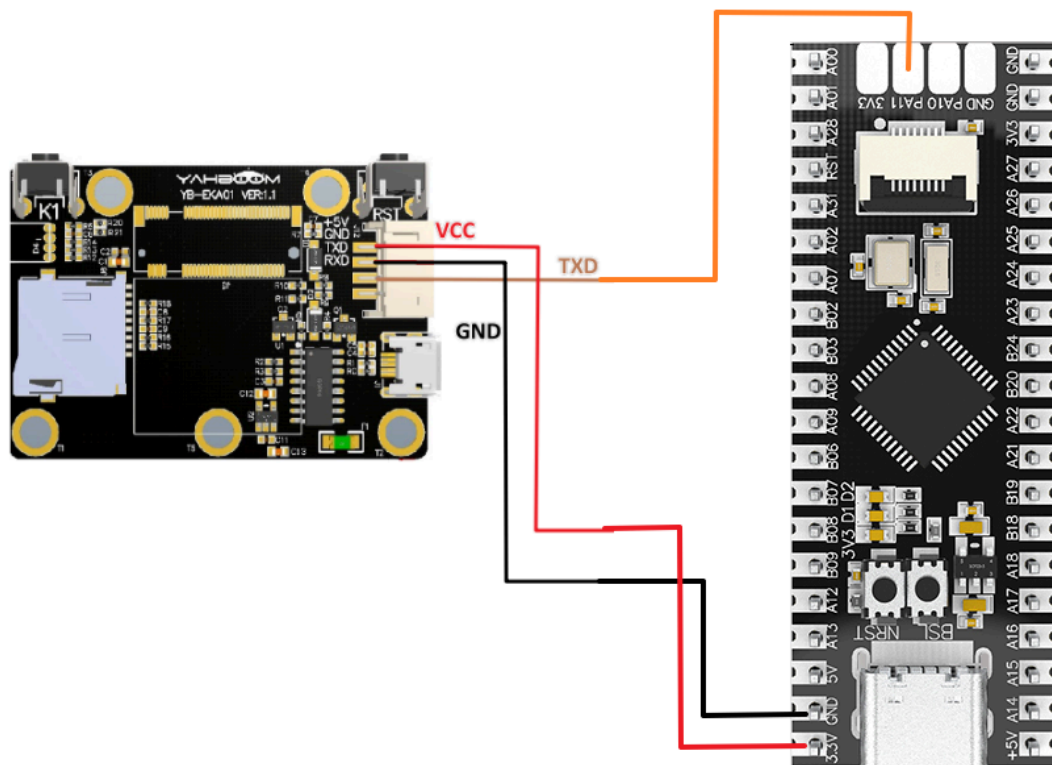
Method 1: Use USB to TTL module

MSPM0G	K210 Vision Module
RX	TXD
GND	GND
VCC	5V

MSPM0G	USB to TTL Module
TX	RXD
GND	GND



Method 2: Use onboard Type-C port



1.3 Main Code Explanation

This example sets the serial port print baud rate to 115200 bps, and the K210 module connection baud rate to 115200 bps.

Type Filter Text... X <<

PROJECT CONFIGURATIO...
Project Config... 1/1 ✓ +

MSPM0 DRIVER LIBRARY ...
SYSTEM (9)
Board 1/1 ✓ +
DMA +
GPIO +
MATHACL +
Configuration NVM +
RTC +
SYSCTL 1/1 ✓ +
SYSTICK 1/1 ✓ +
WWDT +
ANALOG (6)
ADC12 +
COMP +
DAC12 +
GPAMP +
OPA +
VREF +
COMMUNICATIONS (6)
I2C +
I2C - SMBUS +
MCAN +
SPI +
UART 2/4 ✓ +
UART - LIN +
TIMERS (6)
TIMER - CAPTURE +

Software > UART

UART (2 of 4 Added) ⓘ
+ ADD REMOVE ALL
✓ MYUART
✓ K210_Uart
Name MYUART
Selected Peripheral UART0
Quick Profiles
UART Profiles Custom
Basic Configuration
UART Initialization Configuration
Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate 115200
Calculated Baud Rate 115211.52
Calculated Error (%) 0.01
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

Type Filter Text...
X
<<
<
>
Software > UART

PROJECT CONFIGURATION...
Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...
SYSTEM (9)
Board 1/1 ✓ (+)
DMA (+)
GPIO (+)
MATHACL (+)
Configuration NVM (+)
RTC (+)
SYSCTL 1/1 ✓ (+)
SYSTICK 1/1 ✓ (+)
WWDT (+)
ANALOG (6)
ADC12 (+)
COMP (+)
DAC12 (+)
GPAMP (+)
OPA (+)
VREF (+)
COMMUNICATIONS (6)
I2C (+)
I2C - SMBUS (+)
MCAN (+)
SPI (+)
UART 2/4 ✓ (+)
UART - LIN (+)
TIMERS (6)
TIMER - CAPTURE (+)
TIMER - COMPARE (+)

UART (2 of 4 Added) ②

+ ADD
REMOVE ALL

✓ MYUART

✓ K210_Uart

Name K210_Uart
Selected Peripheral UART1

Quick Profiles
UART Profiles Custom

Basic Configuration

UART Initialization Configuration

Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate 115200
Calculated Baud Rate 115211.52
Calculated Error (%) 0.01
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

```

int main(void)
{
    SYSCFG_DL_init();

    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN); // Clear serial interrupt flag
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN); // Enable serial interrupt

    while (1)
    {
        if (k210_msg.class_n != 0) // If class number is not empty
        {
            if(k210_msg.class_n == 2)
            {
                sprintf(buff_com, "x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.x, k210_msg.y, k210_msg.w, k210_msg.h);
                uart0_send_string(buff_com);
            }
        }
    }
}

```

```

        sprintf(buff_com,"str = %s\r\n",k210_msg.msg_msg);
        uart0_send_string(buff_com);
        k210_msg.class_n = 0;
    }

}

delay_ms(500);

}

}

```

Through the above program, if running this example, the k210_msg structure members will have corresponding values and be printed through serial port.

k210_msg: is the structure that receives information. Its main members are:

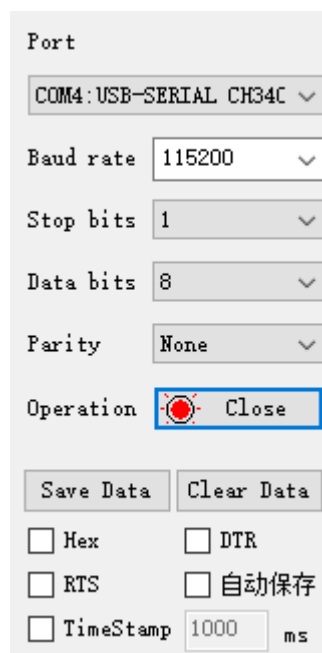
- x: is the x-coordinate of the upper-left corner of the recognized box (range: 0-240)
- y: is the y-coordinate of the upper-left corner of the recognized box (range: 0-320)
- w: is the width of the recognized box (range: 0-240)
- h: is the height of the recognized box (range: 0-320)
- id: is the recognized label
- class_n: class number
- msg_msg[20]: valid data

After data reception function and processing, each member of k210_msg will store valid information. If you want secondary development, you can directly call k210_msg members.

Note: Keil project source code must be placed under SDK path for compilation.

1.4 Experimental Phenomenon

1. After connecting the wires, run the K210 vision module offline [K210 offline running method](#)
2. Set the serial assistant interface as shown in the figure.



3. Then run the barcode recognition example. The serial assistant will print important information transmitted by K210 to MSPM0G, as shown in the figure.

```
x=10, y=23, w=307, h=60
str = 1234567890
x=9, y=22, w=307, h=60
str = 1234567890
x=10, y=21, w=307, h=61
str = 1234567890
x=10, y=22, w=308, h=61
str = 1234567890
x=11, y=21, w=308, h=59
str = 1234567890
x=9, y=19, w=308, h=60
str = 1234567890
x=11, y=21, w=308, h=59
str = 1234567890
x=11, y=18, w=308, h=61
str = 1234567890
x=12, y=19, w=307, h=58
str = 1234567890
x=11, y=19, w=308, h=61
str = 1234567890
x=10, y=17, w=309, h=62
str = 1234567890
x=11, y=18, w=308, h=60
str = 1234567890
x=6, y=13, w=308, h=60
str = 1234567890
x=1, y=7, w=308, h=61
```

Barcode recognition only transmits 5 member variables of k210_msg: x, y, w, h, msg.